SEASUG Paper 048-2026

Wizardry 101: Using a Little GTL Charm on Your SAS Graphs

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ABSTRACT

Have you ever created a graph with an extra legend that even GenAI could not help you remove? This paper explains how Graph Template Language (GTL) can remove the pesky extra legend that SAS SG procedures sometimes create when two different statements are used to produce a graph.

INTRODUCTION

We will create a line graph to visualize fake math test data by class period and time of year, allowing us to assess potential improvement in math scores between the fall and spring semesters.

INITIAL GRAPH

To create our example line graph in SAS, we use PROC SGPLOT with the SERIES and SCATTER statements.

```
proc sgplot data=WORK.mock_data;
series x=period y=mean/group=time;
scatter x=period y=mean/group=time yerrorlower=min
yerrorupper=max;
xaxis label="Class Period";
yaxis label="Average Math Score" values=(0 to 100 by 10);
keylegend/title="Time";
run;
quit;
```

Program 1. Program for Initial Graph

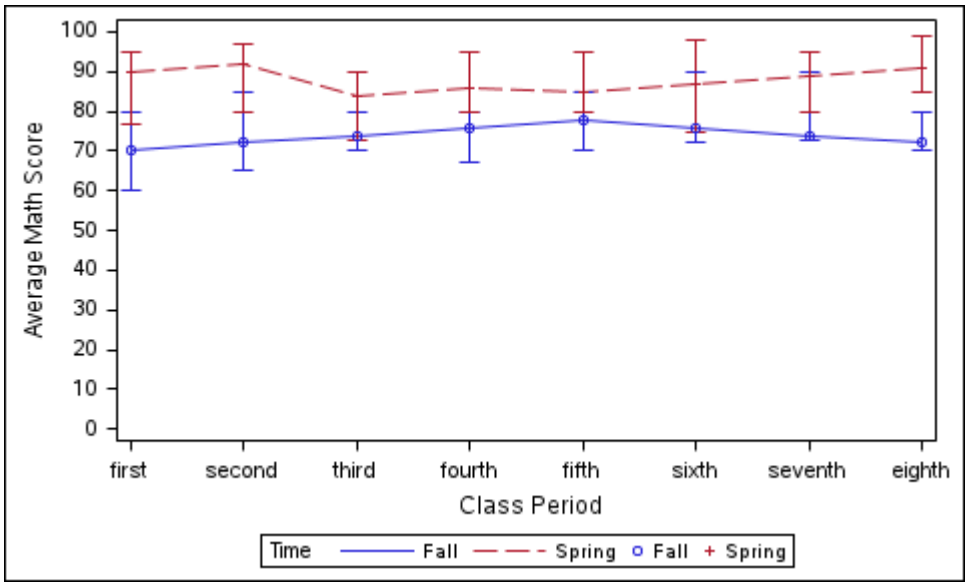


Figure 1. Initial Graph

We notice duplicate information in the legend and ask ourselves, "How can we remove the duplicate information from the legend?"

You go to your favorite GenAI site and ask it:

> I am using SAS to create a line graph with PROC SGPLOT and the SERIES and SCATTER statements, and it is displaying the default legends from both statements. How can I suppress one of the legends?

GenAI responds:

> That's a great question! Suppressing or customizing the legend in PROC SGPLOT is a very common requirement for creating publication-quality graphics. To suppress the legend generated by a specific plot statement (like SERIES or HIGHLOW), you use the NOAUTOLEGEND option within that statement. In your case, if you are using SERIES and SCATTER and you want to keep the legend for, say, the SERIES statement but suppress the one for SCATTER, you would apply NOAUTOLEGEND to the SCATTER statement.

Unfortunately, that solution does not work. After a few minutes, you remember attending an "Ask the Expert" presentation titled "When Do I Use SG Procedures vs. Graph Template Language?" by Michele Ensor. You wonder whether Graph Template Language (GTL) could help remove the duplicate legend information.

USING GRAPH TEMPLATE LANGUAGE (GTL)

We add TMPLOUT="&LINK./data_figure.sas" to our code to output the GTL for the graph. We can then remove the elements we no longer want. In this example, I removed the SCATTER name from the DiscreteLegend line shown below.

```
proc sgplot data=WORK.mock_data
TMPLOUT="&LINK./data_figure.sas";
series x=period y=mean/group=time;
scatter x=period y=mean/group=time yerrorlower=min
yerrorupper=max;
xaxis label="Class Period";
yaxis label="Average Math Score" values=(0 to 100 by 10);
keylegend/title="Time";
run;
quit;
```

Program 2. Program for Outputting the GTL Information

```
proc template;
define statgraph sgplot;
begingraph / collation=binary;
layout overlay / xaxisopts=( Label="Class Period" type=discrete discre
y2axisopts=(labelFitPolicy=Split) yaxisopts=( Label="Average Math Scor
linearopts=( tickvaluelist=( 0 10 20 30 40 50 60 70 80 90 100 ) viewmi
   SeriesPlot X=period Y=mean / Group=time LegendLabel="mean" NAME="SE
   ScatterPlot X=period Y=mean / subpixel=off primary=true Group=time
   DiscreteLegend "SERIES" "SCATTER" / Location=Outside Title="Time";
endlayout;
endgraph;
end;
run;
```

Figure 2. Snippet of Output from Program 2

Finally, we add PROC SGRENDER after our updated PROC TEMPLATE to create the final graph.

```
proc template;
define statgraph sgplot;
begingraph / collation=binary;
layout overlay / xaxisopts=( Label="Class Period" type=discrete discreteopts=(
TickValueFitPolicy=SplitRotate ) )
y2axisopts=(labelFitPolicy=Split) yaxisopts=( Label="Average Math Score"
labelFitPolicy=Split type=linear
linearopts=( tickvaluelist=( 0 10 20 30 40 50 60 70 80 90 100 ) viewmin=0
viewmax=100 ) ) y2axisopts=(labelFitPolicy=Split);
   SeriesPlot X=period Y=mean / Group=time LegendLabel="mean" NAME="SERIES";
   ScatterPlot X=period Y=mean / subpixel=off primary=true Group=time
YErrorUpper=max YErrorLower=min LegendLabel="mean" NAME="SCATTER";
   DiscreteLegend "SERIES" / Location=Outside Title="Time";
endlayout;
endgraph;
end;
run;

proc sgrender data=WORK.mock_data template=sgplot;
run;
quit;
```

Program 3. Program for Creating the Final Graph

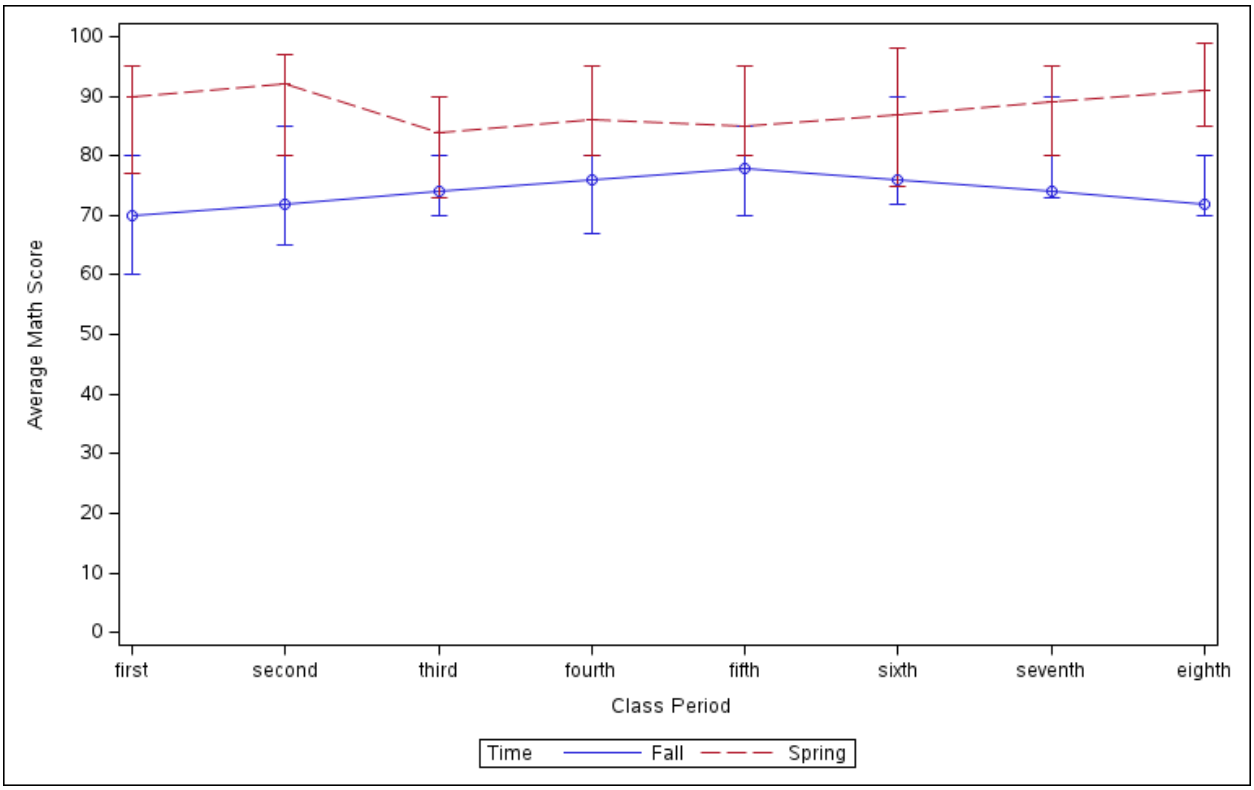


Figure 3. Final Graph

CONCLUSION

I hope this paper helps you understand when GTL can be useful for creating a graph in SAS when the SG procedures do not provide the options needed to achieve the desired final appearance.

REFERENCES

Ensor, Michele. “When Do I Use SG Procedures vs. Graph Template Language?” SAS ASK THE EXPERT. 16 Mar. 2022.

Horstman, Josh, and Richann Watson. "From Muggles to Macros." Research Triangle SAS Users Group. 23 Apr. 2025, Virtual.

ACKNOWLEDGMENTS

This paper is dedicated to my Parents.

CONTACT INFORMATION

Your comments and questions are valued and encouraged. Contact the author at:

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